arXiv:2601.22297v1 [cs.CL] 29 Jan 2026

# Prepare Reasoning Language Models for Multi-Agent Debate with Self-Debate Reinforcement Learning

**Chenxi Liu** [1] **Yanshuo Chen** [1] **Ruibo Chen** [1] **Tianyi Xiong** [1] **Tong Zheng** [1] **Heng Huang** [1]

## Abstract

The reasoning abilities of large language models (LLMs) have been substantially improved by reinforcement learning with verifiable rewards (RLVR). At test time, collaborative reasoning through Multi-Agent Debate (MAD) has emerged as a promising approach for enhancing LLM performance. However, current RLVR methods typically train LLMs to solve problems in isolation, without explicitly preparing them to synthesize and benefit from different rationales that arise during debate. In this work, we propose *Self-Debate Reinforcement Learning* (SDRL), a training framework that equips a single LLM with strong standalone problem-solving ability and the capability to learn from diverse reasoning trajectories in MAD. Given a prompt, SDRL first samples multiple candidate solutions, then constructs a debate context with diverse reasoning paths and generates second-turn responses conditioned on this context. Finally, SDRL jointly optimizes both the initial and debate-conditioned responses, yielding a model that is effective as both a standalone solver and a debate participant. Experiments across multiple base models and reasoning benchmarks show that SDRL improves overall MAD performance while simultaneously strengthening single model reasoning.

## 1. Introduction

Large Language Models (LLMs) have recently demonstrated strong reasoning capabilities, achieving impressive performance on tasks such as mathematics, programming, and tool use (Guo et al., 2025a; Yang et al., 2025; Liu et al., 2025a;d). Reinforcement Learning with Verifiable Rewards (RLVR) (Shao et al., 2024; Guo et al., 2025a; Liu et al., 2025c) plays a crucial role in enabling these capabilities by leveraging tasks with automatically checkable outcomes to provide reliable supervision for scaling reasoning. To further enhance LLM performance, a growing line of work (Wang et al., 2022; Xiong et al., 2025b; Wang et al., 2025b; Zheng et al., 2025b) studies test-time scaling strategies to further improve the reasoning ability of LLMs. Among these approaches, collaborative reasoning (Du et al., 2023; Liang et al., 2024) through Multi-Agent Debate (MAD) has emerged as a promising paradigm: multiple LLM agents propose solutions to a shared question and iteratively update their answers in response to their peers (Choi et al., 2025; Yi et al., 2025).

Despite this promise, effective methodologies for preparing reasoning models for MAD interaction remain underexplored. Existing RLVR methods typically train LLMs to reason and then solve problems within single trajectories, failing to explicitly prepare them for collaborative environments where they must process diverse rationales. This mismatch between training and inference can limit LLM effectiveness in MAD settings (Choi et al., 2025; Kumaran et al., 2025; Li & Goyal, 2025). Furthermore, recent attempts to train debate behaviors typically optimize the entire multi-agent systems, significantly increasing both training costs and deployment complexity (Liao et al., 2025; Liu et al., 2025f; Zhang et al., 2025b; Li et al., 2025). These limitations highlight the need for a framework that produces a *single* LLM capable of both independent problem-solving and productive collaboration in MAD.

In this work, we develop a theoretical analysis that clarifies why debate can help and when its benefits saturate. Inspired by the Dirichlet Compound Multinomial (DCM) framework of Choi et al. (2025), we model multi-agent debate as Bayesian belief updating and recover their key implication that standard MAD induces a martingale over each agent's belief in the correct answer. We then extend this framework by explicitly disentangling the contributions of majority voting and agents' *private critique* in MAD, and show that improving agents' private critique can improve overall MAD performance. Building on this insight, we propose *Self-Debate Reinforcement Learning* (SDRL), a training framework that equips a *single* reasoning model with both strong problem-solving and private critique capa-

[1]Department of Computer Science, University of Maryland, College Park, US. Correspondence to: Heng Huang <>.

*Preprint. February 2, 2026.*

bilities under debate. Given a prompt, SDRL first samples multiple candidate solutions from the current policy and evaluates them using verifiable rewards. SDRL then constructs debate pairs by selecting different candidates from these initial solutions, forcing the model to confront divergent reasoning trajectories. Conditioned on a debate pair, the model generates second-round responses that deliberate over both candidates to produce a final answer. Finally, SDRL jointly optimizes both the initial and second-round responses under verifiable rewards. This joint objective encourages the model to produce accurate first-pass solutions while also learning when and how to revise its reasoning during debate-style interaction, enabling a single trained model to serve as a highly competent agent within MAD.

Overall, our **contributions** can be summarized as follows:

1. We introduce a theoretical framework that disentangles the contributions of debate and majority voting in the MAD paradigm, and we identify improving agents' private critique as a key driver of debate gains.
2. We propose *Self-Debate Reinforcement Learning* (SDRL), a training framework that equips a *single* reasoning model with both independent problem-solving ability and collaborative debate skills by jointly optimizing both objectives.
3. Extensive experiments across multiple reasoning benchmarks demonstrate that SDRL consistently enhances multi-agent debate performance while simultaneously strengthening single-agent accuracy.

## 2. Related Work

**Reinforcement Learning with Verifiable Rewards (RLVR).** Motivated by the success of DeepSeek-R1 (Guo et al., 2025a), a growing body of work adopts RLVR to strengthen the reasoning capabilities of LLMs across a range of domains, including math and STEM problem solving (Wang et al., 2025a; Liu et al., 2025e), search (Jin et al., 2025; Chen et al., 2025a), agentic tool use (Qian et al., 2025; Zhang et al., 2025c), and reward modeling (Chen et al., 2025b; Wang et al., 2025b; Guo et al., 2025b). In parallel, several test-time scaling techniques have been proposed to further improve reasoning performance, such as self-consistency (Wang et al., 2022), self-verification (Zhang et al., 2025a; Liu et al., 2025g), and parallel thinking (Zheng et al., 2025b). However, most existing methods focus on single-agent performance, where each reasoning trajectory is generated by a single policy model, and therefore do not explicitly prepare models for multi-agent debate.

**Multi-Agent Debate (MAD).** Multi-agent systems have proven effective for complex LLM applications that require sophisticated reasoning (Talebirad & Nadiri, 2023; Li et al., 2024a; Han et al., 2024; Xiong et al., 2025a), and several studies propose different MAD settings (Du et al., 2023; Liang et al., 2024; Liu et al., 2024; Smit et al., 2023). However, existing MAD methods do not consistently yield improvements (Smit et al., 2023; Ma et al., 2025) and often fail to outperform strong majority-voting baselines (Choi et al., 2025). Moreover, even state-of-the-art closed-source LLMs can struggle to effectively incorporate conflicting opinions (Kumaran et al., 2025; Li & Goyal, 2025). Recent work attempts to improve MAD with reinforcement learning, but typically either trains an additional aggregator model (Qi et al., 2025; Zhao et al., 2025a) or optimizes an entire multi-agent collaboration system, where each agent has a specialized role (Liao et al., 2025; Liu et al., 2025f; Zhang et al., 2025b; Liu et al., 2025b)s. In contrast, our goal is to train a *single* general-purpose LLM with debate capability—one that remains strong as an individual solver while being explicitly prepared to revise its reasoning when exposed to diverse opinions from other models or users.

## 3. Preliminaries

**Multi-Agent Debate.** We formulate the debate process as an iterative consensus-seeking mechanism among a set of $N$ language model agents, denoted by $\Pi = \{\pi_1, \ldots, \pi_N\}$. Let $\mathcal{Q}$ represent the input space and $\mathcal{O}$ the output space. Each agent $\pi_i$ generates an independent response $o_{i,0} \sim \pi_i(q)$ for a given input $q \in \mathcal{Q}$.

In the *Majority Voting* setting (Zhao et al., 2025b), these independent outputs are immediately aggregated via a voting function $\mathcal{V} : \mathcal{O}^N \to \mathcal{O}$ to produce the final prediction:

$$o_0 = \mathcal{V}(o_{1,0}, \ldots, o_{N,0}). \tag{1}$$

Multi-Agent Debate (MAD) extends this static ensemble approach by introducing a $T$ round communication framework. Under the simultaneous-talk protocol (Chan et al., 2023; Choi et al., 2025), agents update their beliefs in parallel over discrete rounds $t \in \{1, \ldots, T\}$. At step $t$, agent $\pi_i$ conditions its update on the set of peer responses generated in the previous round, formally defined as:

$$\mathcal{O}_i^{(t-1)} = \{o_{j,t-1} \mid j \in \mathcal{N}(i)\}. \tag{2}$$

where $\mathcal{N}(i) \subseteq \{1, \ldots, N\}$ denotes the neighborhood observable to agent $\pi_i$, including itself. The agent then refines its answer using a debate protocol $\mathcal{D}$, which integrates the original query $q$ and the peer context $\mathcal{O}_i^{(t-1)}$ to produce the updated response:

$$o_{i,t} = \mathcal{D}\left(q; \mathcal{O}_i^{(t-1)}\right). \tag{3}$$

This iterative refinement can be viewed as a functional composition over $T$ rounds. The state of agent $\pi_i$ at the final

round is expressed as:

$$o_{i,T} = (\mathcal{D} \circ \cdots \circ \mathcal{D})(q; \mathcal{O}_i) = \mathcal{D}^{(T)}(q; \mathcal{O}_i). \quad (4)$$

Finally, the system-level output is derived by aggregating the refined responses from the terminal state:

$$o_T = \mathcal{V}(o_{1,T}, \ldots, o_{N,T}). \quad (5)$$

Following previous works (Choi et al., 2025; Yi et al., 2025), we focus on homogeneous agent settings, where all $\pi_i$ share identical architectures.

**Reinforcement Learning.** To eschew the computational overhead of a separate critic network, we adopt the Group Relative Policy Optimization (GRPO) framework (Shao et al., 2024; Guo et al., 2025a). For a specific query $q$, the behavior policy $\pi_{\theta_{\text{old}}}$ samples a group of $G$ responses $\{o_i\}_{i=1}^G$. The advantage for the $i$-th response is estimated by normalizing the reward $r_i$ against the group statistics:

$$\hat{A}_{i,t} = \frac{r_i - \text{mean}(\{r_i\}_{i=1}^G)}{\text{std}(\{r_i\}_{i=1}^G)}. \quad (6)$$

Building on this, we employ Decoupled Clip and Dynamic sAmpling Policy Optimization (DAPO) (Yu et al., 2025) to stabilize updates for long chain-of-thought reasoning. DAPO mitigates entropy collapse and reward noise through asymmetric clipping and dynamic sampling constraints. The policy is optimized via the following token-level gradient objective:

$$\begin{aligned} \mathcal{J}_{\text{DAPO}}(\theta) = \; & \mathbb{E}_{(q,o^*)\sim\mathcal{Q},\{o_i\}_{i=1}^G\sim\pi_{\theta_{\text{old}}}(\cdot|q)} \Bigg[ \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \\ & \min\Big( r_{i,t}(\theta)\hat{A}_{i,t},\ \text{clip}\Big(r_{i,t}(\theta), 1-\varepsilon_{\text{low}}, 1+\varepsilon_{\text{high}}\Big)\hat{A}_{i,t} \Big) \Bigg] \\ & \text{s.t.} \quad 0 < \Big| \{ o_i \mid \texttt{is_equivalent}(o^*, o_i) \} \Big| < G, \end{aligned} \quad (7)$$

where $r_{i,t}(\theta) = \frac{\pi_\theta(o_{i,t}|q,o_{i,<t})}{\pi_{\theta_{\text{old}}}(o_{i,t}|q,o_{i,<t})}$ is the importance ratio, and $o^*$ denotes the ground-truth answer.

## 4. Theoretical Analysis

**Motivation.** Under the Dirichlet Compound Multinomial (DCM) model in (Choi et al., 2025), the standard Bayesian debate induces a martingale over each agent's belief in the correct answer, suggesting that debate alone does not improve expected correctness. A key gap is that real agents also apply *private critique* after reading peers, and update their output based on their observation and critique. (Gou et al., 2023) We disentangle this effect with majority voting and show that debate training improves multi-round debate by making this *private critique* positively aligned with the ground truth, thereby inducing a positive drift that breaks martingale neutrality.

**Setup.** Fix an input question $q$ and a finite answer set $\mathcal{A} = \{1, \ldots, K\}$. Without loss of generality, we set answer 1 as the correct one. Following the DCM model (Choi et al., 2025), at round $t$, agent $i$ maintains Dirichlet parameters $\alpha_{i,t} \in \mathbb{R}_+^K$ and observes neighbors $N(i)$. Let $a(\cdot)$ extract the final answer choice from a response. We write $y_{i,t} := a(o_{i,t}) \in \mathcal{A}$ for the discrete answer label used in the DCM analysis. We denote $L^1$-norm as $\|v\|_1 = \sum_{k=1}^K v^{(k)}$ and define the Dirichlet mean as $\bar{\theta}_{i,t} := \alpha_{i,t}/\|\alpha_{i,t}\|_1 \in \Delta^K$.

**Definition 4.1** (Critique-augmented belief update)**.** Let $c_{i,t} \in \mathbb{N}^K$ be the neighbor count vector at round $t$, $c_{i,t}^{(k)} = \sum_{j\in N(i)} \mathbb{1}\{y_{j,t-1} = k\}$. In addition to neighborhood evidence $c_{i,t}$, after observing the round-$t-1$ debate context, agent $i$ computes $\beta_{i,t-1}$ representing its *private critique* from the debate context. The belief update is

$$\alpha_{i,t} \;=\; \alpha_{i,t-1} \;+\; \beta_{i,t-1} \;+\; w_i\, c_{i,t}, \quad (8)$$

where $w_i \geq 0$ controls the strength of the social signal.

**Definition 4.2** (Response generation via DCM)**.** Given $\alpha_{i,t}$, the agent generates $\theta_{i,t} \sim \text{Dirichlet}(\alpha_{i,t})$ and $y_{i,t} \sim \text{Categorical}(\theta_{i,t})$. Marginally, $\Pr(y_{i,t} = k \mid \alpha_{i,t}) = \bar{\theta}_{i,t}^{(k)}$.

Then the belief in the correct answer can be defined via Dirichlet mean:

$$p_{i,t} \;:=\; \bar{\theta}_{i,t}^{(1)} \;=\; \frac{\alpha_{i,t}^{(1)}}{\|\alpha_{i,t}\|_1}. \quad (9)$$

We next let $\mathcal{F}_t$ denote the $\sigma$-algebra of information available at the start of round $t$ (see Appendix E.1 for the explicit definition).

**Definition 4.3** (Advantage of private critique)**.** Assume $\|\beta_{i,t}\|_1 = m_\beta$ is constant (see Appendix E.1 for explanation). The *critique advantage* is

$$\delta_{i,t} \;:=\; \mathbb{E}\Big[\beta_{i,t}^{(1)} \mid \mathcal{F}_t\Big] \;-\; m_\beta\, p_{i,t}. \quad (10)$$

Thus $\delta_{i,t} > 0$ means the critique allocates more belief to the correct answer in this round than a previous round's belief proportional to $p_{i,t}$.

**Theorem 4.4** (Critique induces belief drift)**.** *Assume the mean-consistency condition $\bar{p}_{N(i),t-1} = p_{i,t-1}$ (the same condition under which standard MAD is a martingale in (Choi et al., 2025)). Let $C_i := m_\beta + w_i|N(i)|$, then*

$$\mathbb{E}[p_{i,t} \mid \mathcal{F}_{t-1}] = p_{i,t-1} + \frac{\delta_{i,t-1}}{\|\alpha_{i,t-1}\|_1 + C_i}. \quad (11)$$

*Proof sketch.* Expand $p_{i,t}$ from (8) and take conditional expectation. Use $\mathbb{E}[c_{i,t}^{(1)} \mid \mathcal{F}_{t-1}] = \sum_{j\in N(i)} p_{j,t-1} = |N(i)|\bar{p}_{N(i),t-1}$ and $\alpha_{i,t-1}^{(1)} = p_{i,t-1}\|\alpha_{i,t-1}\|_1$ to obtain Lemma E.1. Under mean-consistency assumption, the neighborhood drift cancels, yielding (11). □